

# Introduction to R Course

## Introductory Lecture

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Wolfgang Viechtbauer

Maastricht University

2026-09-28

# Introduction

- **Wolfgang Viechtbauer**

Department of Psychiatry and Neuropsychology

School for Mental Health and Neuroscience

Maastricht University, The Netherlands

-  <https://www.wvbauer.com>
-  [wolfgang.viechtbauer@maastrichtuniversity.nl](mailto:wolfgang.viechtbauer@maastrichtuniversity.nl)
-  [@wviechtb](#),  [@wviechtb](#),  [@wviechtb](#)

# What is R?

- R is a system for the manipulation, statistical and numerical analysis, and graphical display of data
- freely available under the [GNU General Public License](#) (GPL) → open-source and free/libre
- runs under Windows, macOS, Unix/Linux, ...

# History of S and R

- ... it began May 5, 1976 at:

[Bell Laboratories](#), Murray Hill, New Jersey<sup>1</sup>

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<sup>1</sup>Photo by Alcatel-Lucent Bell Labs

# History of S and R

- informal meeting to discuss development of a new system for statistical computing
- first implementation made by Rick Becker and John Chambers (and a few others)
- called “the system”

*sketch of the system design  
made on the first meeting*

# History of S and R

- “the system” → “S” (the S language)
- first UNIX version of S in 1979 (version 2)
- distributed outside Bell Labs in 1980
- source code released in 1981, then licensed in 1984 for educational and commercial purposes

# History of S and R

- Becker & Chambers (1984). *S: An Interactive Environment for Data Analysis and Graphics*.
- Becker & Chambers (1985). *Extending the S System*.
- Becker, Chambers, & Wilks (1988): *The New S Language: A Programming Environment for Data Analysis and Graphics*.
- Chambers & Hastie (1991). *Statistical Models in S*.
- Chambers (1998). *Programming with Data: A Guide to the S Language*.

# History of S and R

- [S-PLUS](#), a commercial implementation of S, released in 1988 by Statistical Sciences, Inc. (now [TIBCO](#))
- in 1992, [Robert Gentleman](#) and [Ross Ihaka](#) start developing a programming language “not unlike S”

Robert Gentleman and Ross Ihaka<sup>2</sup>

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<sup>2</sup>Photos by Stuart Isett and Kieran Scott

# Some R Milestones

- first binary of R released in 1993
- Ihaka and Gentleman (1996) publish an [article](#) about R in the *Journal of Computational and Graphical Statistics*
- source code released in 1997, [CRAN](#) is started, and the [R Core Team](#) is formed with 9 members
- version 1.0 (2000), 2.0 (2004), 3.0 (2013), 4.0 (2020)
- first [useR! conference](#) in May 2004 in Vienna, Austria
- current version: R 4.6.1 released June 2026

# Other Related Developments

- [Revolution Analytics](#) founded in 2007
- [RStudio](#) founded in 2008 (now called [Posit](#))
- [New York Times article](#) about R in January 2009
- [R Consortium](#) founded in 2015
- [data science](#) develops as a discipline
- [open science](#) / [reproducible research](#) movements
- the emergence of the [tidyverse](#)

# Why is it called R?

- **R**oss Ihaka and **R**obert Gentleman
- pun/play on the name of the S language  
(which in turn was probably a pun based on the C programming language, also developed at Bell Labs)

# Basic Concepts

- command-driven (no point-and-click interface)
- a 'functional' and 'object-oriented' language
- R console: what you see when you start R
- symbol at beginning of line (>): the 'prompt'

# Modes of Interacting with R

- **interactively:** you type commands into the R console line by line and get direct feedback
- **via script files:** you type commands into a script file and then can:
  - copy-paste commands to the console
  - read in and execute all commands at once (e.g., with `source()`, `Rscript`, ...)

# Interactive Mode

- to use R as a “calculator on speed”

```
> x <- c(4,2,3,6)
```

```
> mean(x)
```

```
[1] 3.75
```

- useful for spontaneous exploration of data

# Tab Completion and Command History

- when typing in commands, can use 'tab completion' (esp. useful for long commands)
- type **five** and hit Tab: **five****num** (tada!)
- if ambiguous, can get list with possible options
- type **box** and hit Tab and get list of options
- with  $\uparrow \downarrow$  keys, scroll through command history
- hit ESC (vigorously) if you are 'stuck' somewhere

# Commands Over Multiple Lines

- start typing:  
    `> mean(`
- hit return
- command is syntactically not complete
- continue on next line (prompt is now a `+ sign`)  
    `> mean(  
    + x)`
- hit return  
    `[1] 3.75`

# Always Use Script Files

- promotes:
  - organized programming/analyses
  - code reuse and code sharing
- increases replicability
- easier to fix errors/mistakes
- can write/edit script files with:
  - the built-in editor
  - an external editor
  - an [integrated development environment](#) (IDE)

# The Built-In R Editor

- on **Windows**: rudimentary editor for script files
- on **macOS**: a multidocument editor with 'syntax highlighting' and 'brace-matching'
- start new script: Menu File – New Script
- put cursor in line to be executed and hit Ctrl-R (Windows) or Command-Return (macOS)
- or highlight parts to be executed
- can save/load scripts (usually .r or .R extension)

# External Text Editors

- script files are just plain-text files
- can therefore write them with any text editor
- some editors have functionality for opening multiple documents, code execution, syntax highlighting, brace-matching, and other features
- I personally use [Sublime Text](#)

# Exiting R / Saving the Workspace

- can quit R with:
  - > `quit()`
  - or `q()` or by just closing the window
- prompt: “Save workspace image? [Yes/No/Cancel]”
- if you choose yes: R will save the state of your workspace to the current working directory (into the files `.RData` & `.Rhistory`)
- my recommendation: never do this (choose ‘no’)

# RStudio

- [RStudio](#) is an IDE for R
- free and open source
- runs on Windows, macOS, and Linux
- some of the useful features:
  - syntax highlighting
  - code completion
  - bracket matching
  - object list and command history
  - can run session remotely

# Some RStudio Keyboard Shortcuts

| Description                  | Windows          | macOS               |
|------------------------------|------------------|---------------------|
| Start new script             | Ctrl+Shift+n     | Command+Shift+n     |
| Open script                  | Ctrl+o           | Command+o           |
| Save script                  | Ctrl+s           | Command+s           |
| Close script                 | Ctrl+w           | Command+w           |
| Show keyboard shortcuts      | Alt+Shift+k      | Option+Shift+k      |
| <b>In Script Files:</b>      |                  |                     |
| Run current line / selection | Ctrl+Enter       | Command+Enter       |
| Run entire script file       | Ctrl+Shift+Enter | Command+Shift+Enter |
| Tab completion               | Tab              | Tab                 |
| Show help for function       | F1               | F1                  |

# Working Directory/Folder

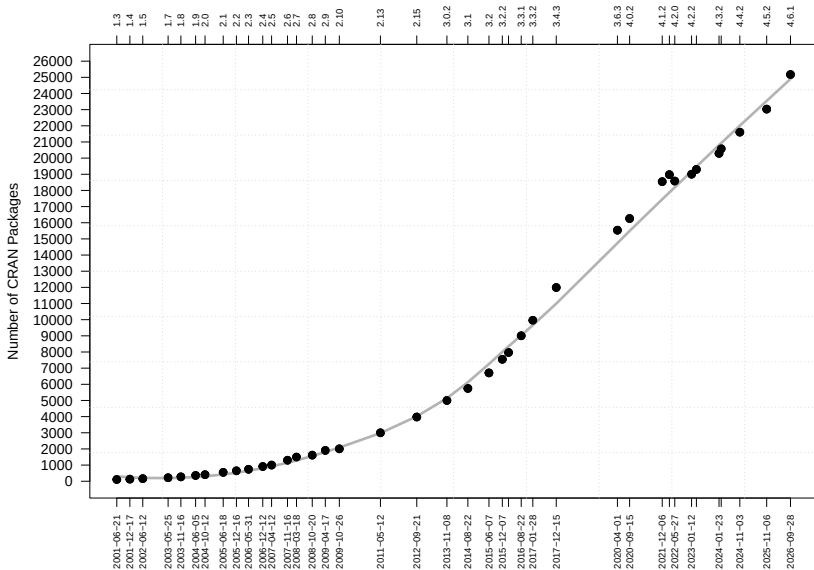
- suppose you have written a script file with the name `rcode.r` and saved it to some directory
- to set the “working directory”, click:
  - **Windows:** Menu File – Change Dir
  - **macOS:** Menu Misc – Change Working Directory
  - **RStudio:** Menu Session – Set Working Directory
- alternatively:
  - `setwd()` – set the working directory
  - `getwd()` – get the current working directory

# R Packages

- an extensive number of add-on “packages” have been contributed by users over the years
- one of the main strengths of R
- many statisticians have adopted R as their primary programming platform → many advanced statistical methods available in R
- [Comprehensive R Archive Network](#) (CRAN): repository for R packages – packages available:

```
> nrow(available.packages())  
[1] 25166
```

# R Packages



# How We Will Proceed ...

- completely hands-on
- will show step-by-step how to do things
- if I go too fast or you have questions, let me know!